Black-box Patch Attack on AutoVLA

Query-only trajectory manipulation on nuScenes front camera inputs

1. **Clean AutoVLA** — decode clean trajectory
2. **Patch placement** — scene-aware + sweep
3. **Black-box search** — evolution / NES, no gradients
4. **Objective** — target-direction trajectory shift
5. **Evaluate** — clean vs attacked trajectory

Attack objective

Use decoded trajectory only: clean prediction vs attacked prediction.

Position-aware loss rewards target-direction lateral shift and final displacement.

Penalizes wrong-way lateral movement and pure forward/backward drift.

Evolution search keeps best candidates because generation outputs are non-smooth.

Scene-aware placement

Heuristic probes construction cues, road/lane texture, and horizon fallback. Black-box sweep then selects the best position per scene.

Results: Smaller Patch Helps, but Still Not Robust

p20 front-camera patch, 5 nuScenes samples, scene-aware position sweep

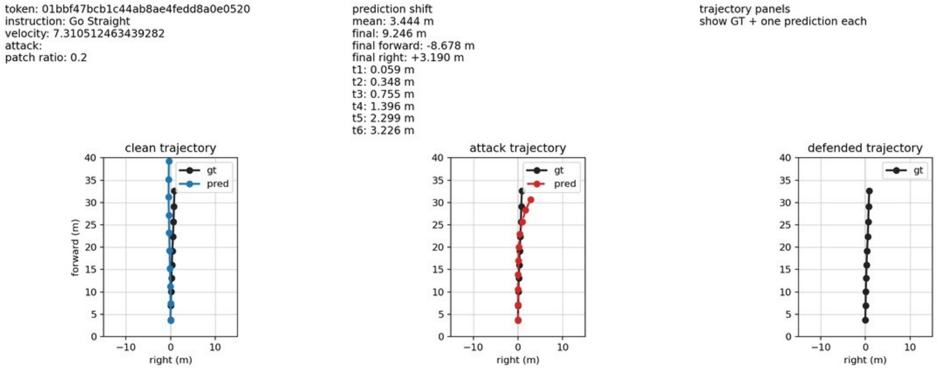
Avg final shift
3.42 m
vs 2.95 m without scene-aware

Avg right shift
0.95 m
vs 0.61 m without scene-aware

Strong cases
2 / 5
final shift > 3 m

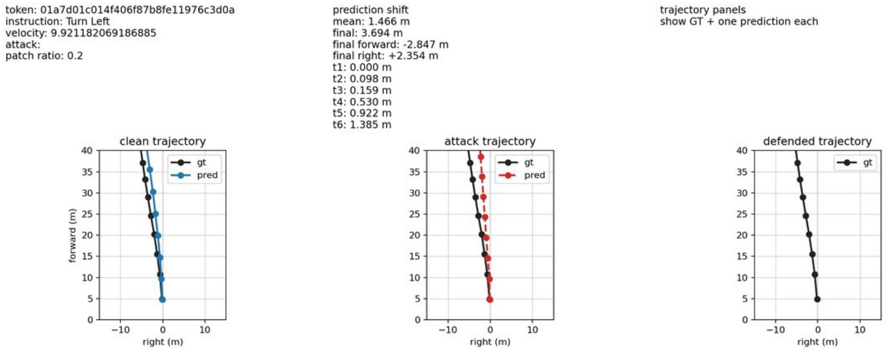
Best case
9.25 m
final shift, 3.19 m right

Best case: horizon-center patch



Token 01bbf47b: final 9.25 m, right +3.19 m

Scene-aware gain: construction cue



Token 01a7d01c: scene-aware improves final 1.31 m -> 3.69 m

Takeaway: scene-aware placement improves targeted lateral shift, but black-box front-only attacks remain inconsistent. Next: object-aware placement for cars, pedestrians, and traffic lights.